Question No : 1 of 50

Mark

Consider a power series $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$ For $x=1$ this series represents

1 / 29

Answer

- e
- e^2
- e^3
- None of them

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Question No : 2 of 50

Marks:

A point x_0 of a differential equation that is not an ordinary point is said to be

1 / 29

Answer

- (•) Non-singular point
- ( ) Singular point
- ( ) Limit point
- ( ) None of them



Page 284

Question No : 3 of 50

Marks: 1 (Time 1

Operator method is the method of the solution of a system of linear homogeneous or linear non-homogeneous differential equations which is based on the process of systemi elimination of the

2 / 29

Answer

- Dependent variables



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- Independent variable
- Choice variable
- None of them

Question No : 4 of 50

Marks: 1 (Time 1

The simple harmonic motion can be drived by combining Newton's 2nd law of motion and .....?

2 / 29

Answer

- Hook's law
- Amplitude
- Newton's 3rd law
- Newton's 1st law

Page 207

Question No : 5 of 50

Marks: 1 (Time 1

Which of the rule in matrices under multiplication does not hold true?

3 / 29

Answer

- Commutative Law
- Associative Law
- Identity Law
- None of these

Page 364

Question No : 6 of 50

Marks: 1 (Time

The vectors $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}$ and $X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$ are

01:58

Answer

- linearly dependent
- linearly independent



Page 398

Question No : 7 of 50

Marks: 1 (Time

The ability of a capacitor to store charge is called _____.

Answer

Power

Capacitance

Page 241

Resistance

Current

Question No : 8 of 50

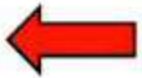
Marks: 1 (Time

Identify the Cauchy-Euler differential equation of order two.

02:19

Answer

- $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+2y=0$



Page 261

- $x\frac{d^2y}{dx^2}+x\frac{dy}{dx}+2y=0$
- $x^2\frac{d^2y}{dx^2}+\frac{dy}{dx}+2y=0$
- $\frac{d^2y}{}+x\frac{dy}{}+2y=0$

Question No : 9 of 50

Marks: 1 (Time

$\det(A - \lambda I) = 0$; this equation is called _____ .

02:44

Answer

- (•) eigen value

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- ( ) characteristic equation
- ( ) eigen vector
- ( ) polynomial equation

Question No : 10 of 50

Marks: 1 (Time

When the characteristic equation has repeated roots, it may _____ to find n linearly independent eigenvectors of the matrix.

02:58

Answer

- possible
- not possible

Page 385

- totally possible
- always happen

Question No : 11 of 50

Marks: 1 (Tim

For finding the general solution of the non-homogeneous system of linear differential equations we need to find _____.

Answer

6 / 29

- complementary solution
- particular solution
- singular solution
- Both complementary and particular solution

Page 401 and page 164

$\frac{dx}{dt} = 5x - 7y + 5Sint$

$\frac{dy}{dt} = 7x + 5y - 11Sint$

03:21

Answer

$X' = \begin{pmatrix} 5 & -7 \\ -7 & -5 \end{pmatrix} X + \begin{pmatrix} 5 \\ 11 \end{pmatrix} Sint$

$X' = \begin{pmatrix} 5 & -7 \\ 7 & -5 \end{pmatrix} X + \begin{pmatrix} 5 \\ 11 \end{pmatrix} Sint$

$X' = \begin{pmatrix} 5 & -7 \\ 7 & -5 \end{pmatrix} X + \begin{pmatrix} 5 \\ -11 \end{pmatrix} Sint$

$X' = \begin{pmatrix} 5 & -7 \\ 7 & 5 \end{pmatrix} X + \begin{pmatrix} 5 \\ -11 \end{pmatrix} Sint$

Question No : 13 of 50

Marks: 1 (T

Let t denote any point in some interval denoted by I. Then the problem of solving the system of differential equations

$$\frac{dX}{dx} = A(t)X + F(t)$$

Subject to the _____ conditions

$X(t_0) = X_0$.

Answer

- dirichlet
- neumann
- initial
- boundary

7 / 29

BSCS.

Question No : 14 of 50

Marks: 1 (Ti

A second order linear differential equation of the form

$x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-8^2)=0$ is called a

Answer

7 / 29

- None of them
- Bessel Equation
- Bernoulli Equation
- Linear equation

Page 311

For the system of differential equations

$\frac{dy}{dt} = 2x, \frac{dx}{dt} = 3y$ the independent variable(s) is (are)



Answer

- y, t
- x, t
- x, y
- t

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} x & y & z & a \\ p & q & r & b \\ l & m & n & o \end{bmatrix}$ then the order of matrix $A \times B$ is

04:20

Answer

- None of them
- 3x3
- 2x3
- 2x4

8 / 29

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If wroskian of the solution vectors X_1 and X_2 is zero, then vectors are ----- .

Answer

- Parallel
- Linearly dependent
- Linearly independent
- None of them



See example of page 113

Question No : 18 of 50

Marks: 1 (

If $\begin{vmatrix} 2-\lambda & -8 \\ 2 & -6-\lambda \end{vmatrix} = 0$ then it gives

Answer

- (●) $\lambda = -6$ of multiplicity 2
- ( ) $\lambda = 2$ of multiplicity 2
- ( ) $\lambda = 6$ of multiplicity 2
- ( ) $\lambda = -2$ of multiplicity 2

Correct option

05:08

Question No : 19 of 50

Marks: 1

The matrix $\begin{bmatrix} \sqrt{3} & 1 & 2 \\ 0 & \sqrt{3} & 1 \\ 0 & 0 & \sqrt{3} \end{bmatrix}$ has eigenvalue of multiplicity

05:33

Answer

- 1
- 2
- 3
- $\sqrt{3}$

The eigenvalues of homogeneous system $X' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} X$, are

05:49

Answer

- $\lambda_1 = i, \lambda_2 = -i$
- $\lambda_1 = 1, \lambda_2 = -1$
- $\lambda_1 = 1, \lambda_2 = -i$
- $\lambda_1 = -1, \lambda_2 = -i$

The matrix $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ has

Answer

- Complex eigen value
- Repeated & real eigen value
- None of them
- Real and unequal value

05:59

If a matrix $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, then eigenvalues are

Answer

- ⊙ 2,-1
- ○ -1,-2
- ○ 1,-2
- ○ 2,1

06:04

If L denote the linear differential operators with constant coefficients,
then $L_1L_4 - L_2L_3$ represents the

06:26

Answer

- $\begin{vmatrix} L_1 & L_3 \\ L_4 & L_2 \end{vmatrix}$
- $\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$
- $\begin{vmatrix} L_1 & L_2 \\ L_4 & L_3 \end{vmatrix}$
- None of these

Which of the following is correct differential operator form of the differential equation $2\frac{dx}{dt} - 5x + \frac{dy}{dt} = e^t$?

06:34

Answer

- $(2D-5)x + Dy = e^{2t}$
- $(2D-5)x + Dy = e^{-t}$
- $(2D-5)x + Dy = e^{t}$
- $(2D+5)x + Dy = e^{t}$

What is the differential operator form of the differential equation
$|a|y'' + by' + 2y = cosx$?

Answer

- $(bD + 2) = cosx$
- $(-aD^2 + bD + 2)y = cosx$
- $(|a|D^2 + bD + 2)y = cosx$
- $(aD^2 + bD + 2)y = cosx$

What is the linear normal form of $y'' - y' + 6y = 0$, by using

$y = x_1$,

$y' = x_1' = x_2$,

$y'' = x_2'$?

06:59

Answer

$x_1 = x_2,\ x_2 = x_2 - 6x$

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$x_1 = x_2,\ x_2' = x_2 + x_1$

$x_2' = x_2$

Answer

$x_1 = x_2$, $x_2 = x_2 - 6x$

$x_1 = x_2$, $x_2' = x_2 + x_1$

$x_2' = x_2$

$x_2' = x_2$, $x_2' = x_3 - 6$



Question No : 27 of 50

What are the eigen values for the matrix $\begin{pmatrix} 1 & 5 \\ 0 & 7 \end{pmatrix}$?

07:14

Answer

- {1,5}
- {1,0}
- {6,5}
- {1,7}



14 / 29

What are the eigen values for the matrix $\begin{pmatrix} 0 & 1 \\ 6 & 7 \end{pmatrix}$?

07:31

Answer

- {0,7}
- {1,6}
- {-1,-6}
- {7,5}

What are the singular points of the differential equation $y'' + xy' + 2y = 0$?

07:44

Answer

- $x = 0$
- $x = 1$
- the set of all real numbers
- does not exist

Question No : 30 of 50

What is the follwing differential equation called?

$(1-x^2)y'' - 2xy' + n(n+1)y = 0$

07:54

Answer

- Cauchy's equation
- Newton's equation
- Legendre's equation
- Bessel's equation



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Question No : 31 of 50

Marks:

Consider the auxiliary equation of the free damped motion $m^2 + 2\lambda m + \omega^2 = 0$ and its roots $m = -\lambda \pm \sqrt{\lambda^2 - \omega^2}$.

If $\lambda^2 - \omega^2 > 0$ then the system is said to be _____.

08:06

Answer

- Under damped
- Over damped
- None of these
- Neither under nor over damped



Page 220

Question No : 32 of 50

The quantity $X = L_r - \frac{1}{C_r}$ is called _____.

Answer

- Resistance of circuit
- Quasi of circuit
- Impedance of circuit
- Reactance of circuit



Page 249

What is annihilator operator of the function $g(x) = 4x^2$?

Answer

- $(D - 4)$
- D^3
- $(D^3 - 4)$
- $(D + 4)$

While solving a 2nd order linear ODE by variation of parameters, we get

08:44

$u_2' = \frac{cosx}{sinx}$, then $u_2 =$ _____.

Answer

- $-\ln |sinx|$
- $-\ln |cosx|$
- $\ln |sinx|$
- $\ln |cosx|$

Wronskian of the functions $y_1(x) = 1, y_2(x) = cosx$ and $y_3(x) = sinx$ is _____.

Answer

- 2
- 3
- 0
- 1

09:04

A singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ said to be a regular singular point if _____.

Answer

- $(x - x_0)Q(x)$ is analytic at $x = x_0$
- $(x - x_0)P(x)$ is analytic at $x = x_0$
- $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at $x = x_0$
- $(x - x_0)^2P(x)$ and $(x - x_0)Q(x)$ are analytic at $x = x_0$

BSCS.

Question No : 37 of 50

$J_{-\frac{2}{3}}(x) - J_{\frac{4}{3}}(x) =$ _____.

Answer

- $2J'_{\frac{4}{3}}(x)$
- $2J'_{\frac{2}{3}}(x)$
- $2J'_{\frac{8}{3}}(x)$
- $2J'_{\frac{1}{3}}(x)$

BSCS.

If $\frac{1}{2}\int_{-1}^{1} P_n(x)P_n(x)(2n+1)dx = 1$ then for $-1 < x < 1$ the Legendre's polynomial is said to be _____.

09:42

Answer

- orthonormal w.r.t. weight function $w(x) = \frac{2n+1}{2}$
- stable w.r.t. weight function $w(x) = \frac{2n+1}{2}$
- orthogonal w.r.t. weight function $w(x) = \frac{2n+1}{2}$



Page 330

- orthogonal w.r.t. weight function $w(x) = 2n+1$

The nature of roots of auxiliary equation deduced from Cauchy Euler equation $4x^2\frac{d^2y}{dx^2} + 8x\frac{dy}{dx} + y = 0$ are .....

Answer

- Real and unequal
- Real and repeated
- None of them
- Complex and repeated

Page 225

Recurrence relation of the Differential equation $y'' + y = 0$, is .....

Answer

$c_{k+1} = \frac{1}{k(k+1)} c_k$

$c_{k+2} = \frac{1}{(k+1)(k+2)} c_k$

$c_{k+1} = -\frac{1}{2} c_k$

Answer

$c_{k+1} = \frac{1}{k(k+1)} c_k$

$c_{k+2} = \frac{1}{(k+1)(k+2)} c_k$

$c_{k+1} = -\frac{1}{2} c_k$

$c_{k+2} = -\frac{1}{(k+1)(k+2)} c_k$



In the equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ differentiate ordinary and singular points with respect to analytic function?

Answer

Reduce the third - order equation $\frac{d^3y}{dx^3}+y+2\frac{dy}{dx}-\frac{d^2y}{dx^2}+e^x=0$ to the normal form.

Answer

Find the Eigen values of the given matrix

$A = \begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix}$.

Answer

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Question No : 44 of 50

Marks: 3 (Time 6 Min

A mass weighing 2lb stretches a spring to 6 inches. Determine the equation of the simple harmonic motion.

11:36

Answer

Styles | Format | Font | Size

Question No : 45 of 50

Show that $y = c_1 x^{-1} + c_2 x^4$ is a solution of $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} - 4y = 0.$

Answer



Question No : 46 of 50

Marks: 3 (Time 8 Min)

Define regular singular points of the equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, where coefficients of the differential equation are polynomials with no common factors.

Answer

Styles | Format | Font | Size

Show that vectors $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-6t}$ and $X_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t}$ are linearly dependent.

Answer

Styles Format Font Size

See the related example on

Page 393

13:12

Question No : 48 of 50

Find the interval of convergence of the given power series

$$\sum_{n=0}^{\infty} \frac{(x+7)^n}{\sqrt{n}}$$

Answer

B I U Styles Format Font Size A

See the related problem on

Page 271

Question No : 49 of 50

Find the general solution of the given differential equation on $(0, \infty)$

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1)y = 0$$

Answer



See related problem on

Page 314

Question No : 50 of 50

Write the solution $x(t) = \frac{4}{3}\cos 3t - \frac{5}{3}\sin 3t$ of an initial value problem in the form $x(t) = A\sin(\omega t + \varphi)$.

Answer

B I U | Styles | Format | Font | Size | A

See related problem on

Page 215

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